

Mains electricity and the National Grid

Electricity
Paper 1 · Combined and Separate
32 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 15 starred (★) questions first.

	QUESTION	ANSWER	✓
1	What do a.c. and d.c. stand for?	Alternating current and direct current	☐
2★	What is an alternating current (a.c.)?	A current that repeatedly reverses direction.	☐
3★	State the frequency and potential difference of the UK mains electricity supply.	50 Hz and about 230 V	☐
4★	State the colour of the insulation on the live, neutral and earth wires.	Live: brown Neutral: blue Earth: green and yellow stripes	☐
5★	State the function of the live wire in a three-core cable.	It carries the alternating potential difference from the supply.	☐
6★	State the function of the neutral wire in a three-core cable.	It completes the circuit.	☐
7★	State the function of the earth wire in a three-core cable.	It is a safety wire that stops the appliance becoming live.	☐
8	State the potential difference between the live wire and the earth wire.	About 230 V	☐
9★	Name the wires in a three-core cable that are at, or close to, 0 V.	The earth wire and the neutral wire	☐
10	State when the earth wire will carry a current.	Only when there is a fault.	☐
11★	In a plug, name the wire that is connected to the fuse.	The live wire	☐
12★	Explain why touching the live wire is dangerous even when the appliance is switched off.	The live wire is still at about 230 V, so a current would flow through the person to earth.	☐

	QUESTION	ANSWER	✓
13	Define power in terms of energy.	The rate at which energy is transferred.	☐
14 ★	State the standard unit of power.	Watt (W)	☐
15	Give two things that affect the energy transferred by an appliance.	Its power and the time it is switched on	☐
16	What does a higher power rating tell you about an appliance?	It transfers more energy each second.	☐
17 ★	Describe the National Grid and its purpose.	A system of cables and transformers linking power stations to consumers.	☐
18	State what is transferred across the National Grid.	Electrical power	☐
19	State the function of transformers.	To increase or decrease the potential difference.	☐
20	What does a step-up transformer do to the potential difference?	It increases it.	☐
21 ★	What does a step-up transformer do to the current?	It decreases it.	☐
22 ★	Explain why step-up transformers are used before electricity is transmitted.	A higher potential difference means a smaller current, so less energy is wasted heating the cables.	☐
23 ★	What is direct current (d.c.)?	A current that flows in one direction only.	☐
24	Name a source of direct current.	A cell or battery	☐
25	State the equation linking power, potential difference and current.	power = potential difference $\times$ current ($P = V I$)	☐
26	State the equation linking power, current and resistance.	power = current ² $\times$ resistance ($P = I^2 R$)	☐
27	State the equation linking energy transferred, power and time.	energy transferred = power $\times$ time ($E = P t$)	☐
28	State the equation linking energy transferred, charge flow and potential difference.	energy transferred = charge flow $\times$ potential difference ($E = Q V$)	☐

QUESTION		ANSWER	✓
29	State the equation for a transformer that is 100% efficient.	p.d. × current in the secondary = p.d. × current in the primary ($V_s \times I_s = V_p \times I_p$)	☐
30	A battery-powered fan is switched on. Describe the useful energy transfer.	Energy is transferred electrically from the chemical store of the battery to the kinetic store of the fan.	☐
31 ★	Explain why step-down transformers are used before electricity reaches homes.	To reduce the potential difference to a safer value (230 V).	☐
32	Explain why any connection between the live wire and earth is dangerous.	A very large current would flow, which could cause a fire or a fatal shock.	☐